Sri Harsha Mudumba

sriharshamudumba@gmail.com | +1 (515) 916-3011 | Ames, IA | [LinkedIn] | [GitHub]

SUMMARY

ML engineer who works directly with enterprise clients to scope production problems and build the systems that solve them, across **telecom, biotech, and fintech** domains. At Procal Technologies, collaborates directly with telecom service provider clients (**T-Mobile, Verizon, Motorola**) to understand their needs and ship products suited to them, including a **post-training stack**, a distilled **31B-to-4B production model**, and a **35% p95 latency reduction** in the inference stack. RL research background (PPO, Q-learning, reward shaping) from graduate research on hardware-software co-design.

TECHNICAL SKILLS

Applied ML: Client requirements gathering, domain-adaptive fine-tuning, production ML system design across telecom, biotech, and fintech clients

Post-Training & Alignment: SFT, LoRA, QLoRA, PEFT, PPO, reward engineering, knowledge distillation, LLM-as-judge evaluation, dataset curation, Hugging Face TRL and PEFT

Distributed Training: PyTorch DDP, FSDP, DeepSpeed, multi-node A100 clusters, mixed-precision training, training stability debugging

Inference & Runtime: vLLM, custom fused CUDA kernels, C++ operator fusion, continuous batching, KV-cache management, quantization (PTQ, QAT)

Infrastructure: Python, C++, CUDA, FastAPI, Docker, Kubernetes, AWS (EC2, S3, SageMaker), MLflow, CI/CD for ML

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Procal Technologies | Remote, US **Oct 2025 - Present**

- Works directly with telecom service provider clients including **T-Mobile, Verizon, and Motorola** to understand their operational needs and build production ML systems around them.
- Built the post-training stack for TelecomLM based on client requirements, running domain-adaptive continued pre-training and SFT with LoRA over **135,000+ curated QA pairs** spanning five telecom knowledge domains.
- Distilled a Gemma 4 31B teacher into a 4B student and shipped a production **LLM-as-judge** pipeline that uses the 31B model to evaluate fine-tuned 4B outputs, replacing manual review as the release quality gate.
- Wrote custom fused CUDA kernels and restructured C++ operator fusion and async scheduling in the inference stack, cutting **p95 latency 35%** under concurrent load, and deployed serving with vLLM.
- Built a benchmarking harness with A/B experimentation and automated rollback triggers that stress-tests model variants before release, catching **two silent regressions** before they reached production.

ML Research | Iowa State University | Ames, IA **Aug 2023 - Aug 2025**

- Designed multi-objective reward shaping for PPO and Q-learning agents across accuracy, latency, and RRAM endurance constraints, achieving **86-88% inference energy savings** on ResNet-50 within 1% accuracy.
- Integrated PyTorch with CIMLOOP to translate hardware simulation state into training signals, letting RL agents adapt exit policy in real time; PPO outperformed the fixed-exit baseline **1.9x**, extending projected device lifetime to **~1.02x10^7 years**.
- Trained PPO and Q-learning agents on multi-node A100 clusters using PyTorch DDP with mixed-precision training, debugging training stability issues across distributed runs.
- Built custom CUDA kernels for per-layer inference profiling to measure per-layer energy and latency cost, feeding those measurements into the reward signal for the RL agents.

Software Engineer | Cognizant Technology Solutions | Bengaluru, India **Aug 2020 - Jul 2023**

- Worked directly on client accounts including **BioMarin and Intuit**, handling data pipeline issues affecting their downstream ML training workflows across different industry domains.
- Traced silent data quality failures across distributed ETL stages feeding ML training pipelines from **5+ TB** production databases and implemented validation checkpoints that restored SLA-compliant delivery.
- Rewrote critical data processing paths that were stalling ML feature pipelines, cutting processing latency **30%** and unblocking downstream training throughput.

EDUCATION

M.S. Computer Engineering | Iowa State University | Ames, IA **Aug 2023 - Aug 2025**

B.Tech Electronics & Communication Engineering | Amrita Vishwa Vidyapeetham **Aug 2016 - Aug 2020**